12 General Renormalization Theory

12.3 Is Renormalizability Necessary?

In the previous section we found a special class of theories having only a finite number of terms in the Lagrangian, to which the renormalization program is nevertheless applicable. These are theories in which all interactions satisfy the renormalizability condition

$$\Delta_i \equiv 4 - d_i - \sum_f n_{if}(s_f + 1) \geq 0,$$

where d_i and n_{if} are the numbers of derivatives and fields of type f in interactions of type i, and s_f is (with some qualifications) the spin of fields of type f. For renormalization to work in such theories, it is also usually necessary that all renormalizable interactions that are allowed by symmetry principles should actually appear in the Lagrangian.

It is important that there are only a limited number of such interaction types. Δ_i becomes negative if we have too many fields or derivatives, or fields of too high spin. Barring special cancellations, there are no renormalizable interactions at all involving fields with $s_f \geq 1$, because the only possible term in the Lagrangian with $\Delta_i \geq 0$ that involves such a field along with two or more other fields would involve a single $s_f = 1$ field along with two scalars and no derivatives, which would not be Lorentz-invariant. We shall see in Volume II that general, massless, spin one gauge fields in a suitable gauge effectively have $s_f = 0$, like the photon. Also, in Volume II we shall see that even massive gauge fields may effectively have $s_f = 0$, depending on where their mass comes from. Leaving aside these special cases, Table 12.2 gives a list of *all* renormalizable terms in the Lagrangian density that are allowed by Lorentz invariance and gauge invariance involving scalars ($s = 0$), photons ($s = 0$), and spin $\frac{1}{2}$ fermions ($s = \frac{1}{2}$).

We see that the requirement of renormalizability puts severe restrictions on the variety of physical theories that we may consider. Such restrictions provide a valuable key to the structure of physical theories. For instance, Lorentz and gauge invariance by themselves would allow the introduction of a 'Pauli' term proportional to $\bar{\psi}[\gamma_\mu, \gamma_\nu]\psi F^{\mu\nu}$ in the Lagrangian of quantum electrodynamics, which would make the magnetic moment of the electron an adjustable parameter, but we exclude such terms because they are not renormalizable. The successful predictions of quantum electrodynamics, such as the calculation of the magnetic moment of the electron outlined in Section 11.3, may be regarded as validations of the principle of renormalizability. The same applies to the standard model of weak, electromagnetic, and strong interactions, to be discussed in Volume II; there are any number of terms that might be added to this theory, such as four-fermion interactions among quarks and leptons, that would invalidate all the predictions of the standard model, and are excluded only because they are non-renormalizable.

Must we believe that the Lagrangian is restricted to contain only renormalizable interactions? As we saw in the previous section, if we include in the Lagrangian *all* of the infinite number of interactions allowed by symmetries, then there will be a counterterm available to cancel every ultraviolet divergence. In this sense, as said earlier, non-renormalizable theo-

Table 12.2. Allowed renormalizable terms in a Lagrangian density involving scalars ϕ, Dirac fields ψ, and photon fields A^μ. Here n_{if} and d_i are the number of fields of type f and the number of derivatives in an interaction of type i, and Δ_i is the dimensionality of the associated coefficient.

	n_{if}		d_i	Δ_i	$\mathcal{H}_i$
Scalars	Photons	Spin $\frac{1}{2}$			
1	0	0	0	3	ϕ
2	0	0	0	2	ϕ^2
2	0	0	2	0	$\partial_\mu\phi\partial^\mu\phi$
3	0	0	0	1	ϕ^3
4	0	0	0	0	ϕ^4
2	1	0	1	0	$\phi\partial_\mu\phi A^\mu$
2	2	0	0	0	$\phi^2 A_\mu A^\mu$
1	0	2	0	0	$\phi\bar{\psi}\psi$
0	2	0	2	0	$F_{\mu\nu}F^{\mu\nu}$
0	0	2	0	1	$\bar{\psi}\psi$
0	0	2	1	0	$\bar{\psi}\gamma^\mu\partial_\mu\psi$
0	1	2	0	0	$\bar{\psi}\gamma^\mu A_\mu\psi$

ries are just as renormalizable as renormalizable theories, as long as we include all possible terms in the Lagrangian.

In recent years it has become increasingly apparent that renormalizability is not a fundamental physical requirement, and that in fact any realistic quantum field theory will contain non-renormalizable as well as renormalizable terms. This change in point of view can be traced in part to the continued failure to find a renormalizable theory of gravitation. In the general class of metric theories of gravitation governed by Einstein's principle of equivalence there are no renormalizable interactions at all—generally covariant interactions must be constructed from the curvature tensor and its generally covariant derivatives, and hence, even in a 'gauge' where the graviton propagator goes as k^{-2}, these interactions involve too many derivatives of the metric for renormalizability. In particular, we can easily see that general relativity is non-renormalizable from the fact that its coupling constant $8\pi G_N = (2.43 \cdot 10^{18}\ \text{GeV})^{-2}$ has negative dimensionality. Even if nothing else did, the cancellation of divergences due to virtual gravitons would require that the Lagrangian contain all interactions allowed by symmetries—not only interactions involving gravitons, but involving any particles.

But if renormalizability is not a fundamental physical principle, then how do we explain the success of renormalizable theories like quantum electrodynamics and the standard model? The answer can be seen by simple dimensional analysis. We have already noted that the coupling constant of an interaction of type i has dimensionality

$$[g_i] \sim [\text{mass}]^{\Delta_i}, \tag{12.3.1}$$

where Δ_i is the index (12.1.9). Non-renormalizable interactions are just those whose coupling constants have the dimensionality of *negative* powers of mass. Now, it is not unreasonable to guess from (12.3.1) that the coupling constants not only have dimensionalities governed by Δ_i, but are roughly of order

$$g_i \approx M^{\Delta_i}, \tag{12.3.2}$$

where M is some common mass. (This is found to be actually the case in the effective field theories discussed below and in more detail in Volume II.) In calculating physical processes at a characteristic momentum scale $k \ll M$, the inclusion of a non-renormalizable interaction of type i with $\Delta_i < 0$ will introduce a factor $g_i \approx M^{\Delta_i}$, which on dimensional grounds must be accompanied by a factor $k^{-\Delta_i}$, and so the effect of such an interaction is suppressed[1] for $k \ll M$ by a factor $(k/M)^{-\Delta_i} \ll 1$. (This argument will be made more carefully using the method of the renormalization group in Volume II.) The success of the renormalizable theories of electroweak and strong energies shows only that M is very much larger than the energy scale at which these theories have been tested.

For instance, the leading non-renormalizable corrections to the conventional electrodynamics of electrons or muons would be those interactions of dimension 5, which are suppressed by only one factor of $1/M$. There is just one such interaction allowed by Lorentz, gauge, and CP invariance, a Pauli term of order $(ie/2M)\bar{\psi}[\gamma_\mu, \gamma_\nu]\psi F^{\mu\nu}$. According to Eqs. (10.6.24), (10.6.17), and (10.6.19), such a term would contribute an amount of order $4e/M$ to the magnetic moment of the electron or muon. The calculated value of the magnetic moment of the electron agrees with experiment to within terms of order $10^{-10}e/2m_e$, so M must be greater than about $8 \cdot 10^{10} m_e = 4 \cdot 10^7$ GeV.

This limit may be weakened if other symmetries restrict the form of the non-renormalizable interactions. For instance, the conventional Lagrangian of quantum electrodynamics is invariant under a chiral transformation $\psi \to \gamma_5\psi$, except for a change of sign of the fermion mass term $-m\bar{\psi}\psi$. If we assume that the full Lagrangian is invariant under a formal symmetry $\psi \to \gamma_5\psi$, $m \to -m$, then a Pauli term in the Lagrangian would have to appear with an extra factor m/M, so that its contribution to the magnetic moment would be only of order $4em/M^2$. Because of the extra factor of m, here it is the muon rather than the electron that provides the most useful limit on M. The calculated value of the magnetic moment of the muon agrees with experiment to within terms of order $10^{-8}e/2m_e$, so M must be greater than about $\sqrt{8 \cdot 10^8}\, m_\mu = 3 \cdot 10^3$ GeV. In any case, if M is anywhere near as large as 10^{18} GeV, then we are certainly justified in neglecting any non-renormalizable interactions that might appear in quantum electrodynamics.

[1] It is essential at this point to assume that the ultraviolet divergences have been removed by renormalization, so that there are no factors of an ultraviolet cutoff Λ to mess up our dimensional analysis. Otherwise, dimensional analysis tells us that for $\Lambda \to \infty$, each additional non-renormalizable coupling constant factor g_i with $\Delta_i < 0$ would be accompanied with a growing factor $\Lambda^{-\Delta_i}$. This dimensional argument led Heisenberg [9] very early to classify interactions according to the dimensionality of their coupling constants, and to suggest [10] that new effects might arise at energies of order g_i^{1/Δ_i}, as for instance at the energy $G_F^{-1/2} \approx 300$ GeV, where G_F is the four-fermion coupling constant of the Fermi beta decay theory. After the development of renormalization theory it was noted by Sakata *et al.* [11] that the non-renormalizable theories are those whose coupling constants have negative dimensionality.

These considerations help us to cope with some of the puzzles associated with higher-derivative terms in the Lagrangian. For instance, in the general theory of a real scalar field ϕ, we would expect to find terms in the Lagrangian density of the form $\phi\Box^n\phi$. Any one such term would make a direct contribution to the scalar self-energy function $\Pi^*(q^2)$ proportional to $(q^2)^n$. If we were to include this contribution to all orders, but ignore all other effects of non-renormalizable interactions, then the propagator $\Delta'(q^2) = 1/(q^2 + m^2 - \Pi^*(q^2))$ would not have the simple pole in q^2 at negative q^2 expected from the general arguments of Section 10.7, but n such poles (some of which may coincide), generally at complex values of q^2. But if the non-renormalizable term $\phi\Box^n\phi$ has a coefficient of order $M^{-2(n-1)}$, where $M \gg m$, then the extra poles are at q^2 of order M^2, where it is illegitimate to ignore the infinite number of other non-renormalizable interactions that must also appear in the Lagrangian. Thus the appearance of higher-derivative terms in a general non-renormalizable Lagrangian is not in conflict with the general principles underlying quantum field theory that were used in Section 10.7. But by the same token, we also cannot use higher-derivative terms to avoid ultraviolet divergences altogether, as has been repeatedly proposed. A term $M^{-2(n-1)}\phi\Box^n\phi$ in the Lagrangian density provides a cutoff at momenta $q^2 \approx M^2$, but at these momenta we cannot ignore all the other non-renormalizable interactions that must be present.

Although highly suppressed, non-renormalizable interactions may be detectable if they have effects that would otherwise be forbidden. For instance, we will see in Section 12.5 that the symmetries of charge-conjugation and space-inversion invariance are an automatic consequence of the structure of the electromagnetic interactions that is imposed by gauge invariance, Lorentz invariance, and renormalizability, but we can easily imagine non-renormalizable terms that would violate these symmetries, such as an electron electric dipole moment term $\bar{\psi}\gamma_5[\gamma_\mu, \gamma_\nu]\psi F^{\mu\nu}$, or the Fermi interaction $\bar{\psi}\gamma_5\gamma_\mu\psi\bar{\psi}\gamma^\mu\psi$. It is widely believed today that the conservation of baryon and lepton number is violated by very small effects of highly suppressed non-renormalizable interactions. Another example of a detectable non-renormalizable interaction is provided by gravitation. As mentioned before, gravitons have no renormalizable interactions at all. But, of course, we detect gravitation, because it has the special property that the gravitational fields of all the particles in a macroscopic body add up coherently.

Although non-renormalizable theories involve an infinite number of free parameters, they retain considerable predictive power: [12] they allow us to calculate the non-analytic parts of Feynman amplitudes, like the $\ln q$ and $q \ln q$ terms in the one-dimensional examples at the beginning of the previous section. Such calculations just reproduce the results required by the axiom of S-matrix theory, that the S-matrix has only those singularities required by unitarity.

Paradoxically, it is just in the case where symmetry principles forbid renormalizable interactions that non-renormalizable quantum field theories prove the most useful. In such cases we can derive a useful perturbation theory by expanding in powers of k/M. This has been worked out in detail for the theory of low-energy pions [12,13], to be discussed in detail in Volume II, and the theory of low-energy gravitons [14]. For a simpler example, consider the theory of a real scalar field, satisfying the principle of invariance under the field translation

$$\phi(x) \longrightarrow \phi(x) + \epsilon$$

with ϵ an arbitrary constant. This symmetry forbids any renormalizable interactions or scalar mass, but it allows an infinite number of non-renormalizable derivative interactions

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{g}{4}(\partial_\mu\phi\partial^\mu\phi)^2 - \cdots ;$$

where $g \approx M^{-4}$, and '$\cdots$' denotes terms with more derivatives or fields. (For simplicity, it is assumed here that the theory also has a symmetry under the reflection $\phi \to -\phi$.) According to the above dimensional analysis, the graph for a general reaction in which all energies and momenta are of order $k \ll M$ is suppressed by a factor $(k/M)^\nu$, where

$$\nu = -\sum_i V_i \Delta_i = \sum_i V_i(d_i + n_i - 4),$$

with n_i and d_i the numbers of scalar fields and derivatives in an interaction of type i, and V_i the number of vertices for these interactions in our graph. For $k \ll M$, the dominant contributions to any process are those with the smallest value of ν. The formula for ν can be put in a more useful form by using the familiar topological identities for a connected graph:

$$\sum_i V_i = I - L + 1, \qquad \sum_i V_i n_i = 2I + E,$$

where I, E, and L are the numbers of internal lines, external lines, and loops in our graph. Combining these relations gives

$$\nu = 2E - 4 + 4L + \sum_i V_i(d_i - n_i),$$

Now, the field translation symmetry requires that every field must be accompanied with at least one derivative, so the quantity $d_i - n_i$ as well as L is non-negative for all interactions. Thus for a given process (that is, a fixed number E of external lines) the dominant terms will be those constructed solely from *tree* graphs (i.e., $L = 0$), and interactions with the minimum number $d_i = n_i$ of derivatives. That is, in leading order we can take the Lagrangian density to depend only on *first* derivatives of the field. Higher-order corrections may involve loops and/or interactions with more derivatives on some fields. But to any given order ν in k/M, we need only consider a finite number of graphs, those with $L \leq (4 - 2E + \nu)/4$, and only a finite number of interaction types.

For instance, scalar–scalar scattering is given in leading order by the one-vertex tree graph calculated using the interaction $-g(\partial_\mu\phi\partial^\mu\phi)^2$ in first order. According to our formula for ν, the leading correction, suppressed at low energy by a factor $(k/M)^2$, arises from another single-vertex tree graph, produced by an interaction with two additional derivatives of the form[2] $\partial_\mu\partial_\nu\phi\partial^\mu\partial^\nu\phi\partial_\rho\phi\partial^\rho\phi$. The next corrections, suppressed at low energy by two further factors of k/M, arise both from the one-loop diagram of Figure 12.4 (including

[2]In accordance with the remarks of Section 7.7, we are excluding interactions involving $\Box\phi$, because the field equation for ϕ can be used to express such interactions in terms of the others.

permutations of external lines), calculated using only the interaction $-g(\partial_\mu\phi\partial^\mu\phi)^2$, and also from tree graphs with a single vertex arising from a quartic interaction with eight derivatives, whose couplings contain infinite parts that cancel the ultraviolet divergence from the loop graph.[3] The loop graph also yields finite terms in the scattering amplitude proportional to terms like $s^4 \ln s + t^4 \ln t + u^4 \ln u$, $s^2t^2 \ln u + t^2u^2 \ln s + u^2s^2 \ln t$, etc., with calculable coefficients proportional to g^2. These finite terms simply represent the correction to the lowest-order scattering amplitude needed to ensure the unitarity of the S-matrix, but perturbative quantum field theory is by far the easiest way of calculating them.

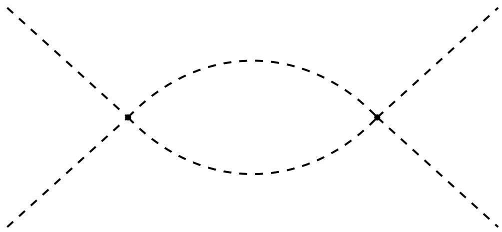

Figure 12.4. One-loop diagram for scalar–scalar scattering in the theory with derivative quadrilinear interactions.

Although non-renormalizable theories can provide useful expansions in powers of energy, they inevitably lose all predictive power at energies of the order of the common mass scale M that characterizes the various couplings. If we were to take these expansions literally, the results for S-matrix elements would violate unitarity bounds for $E \gg M$. There seem to be just two possibilities about what happens at such energies. One is that the growing strength of the effects of the non-renormalizable couplings somehow saturates, avoiding any conflict with unitarity [15]. The other is that new physics of some sort enters at the scale M. In this case, the non-renormalizable theories that describe nature at energies $E \ll M$ are just *effective field theories* rather than truly fundamental theories.

Probably the earliest example of an effective field theory was derived in the 1930s by Euler *et al.* [16], as a theory of low-energy photon–photon interactions. (See Section 1.3.) In effect, they calculated the contribution to photon–photon scattering of Feynman diagrams such as Figure 12.5, and found that at energies much less than m_e the scattering of light by light was the same as would be calculated with an effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{2\alpha^2}{45m_e^4}\left[(\mathbf{E}^2 - \mathbf{B}^2)^2 + 7(\mathbf{E}\cdot\mathbf{B})^2\right] \\ + \text{higher orders in } \frac{eE}{m_e^2} \;\&\; \frac{eB}{m_e^2}$$

[3]These are the only ultraviolet divergences encountered in one-loop graphs if we use dimensional regularization. For other methods of regularization there are also quartic and quadratic divergences, which are cancelled by counterterms in the four-scalar interactions with four or six derivatives.

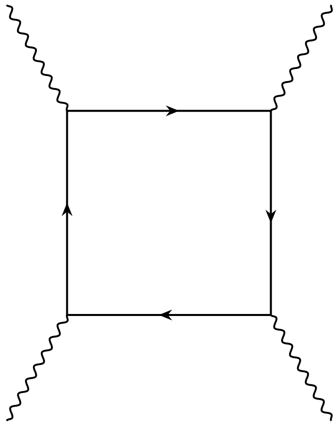

Figure 12.5. Diagram for photon–photon scattering, whose effect at low energy can be calculated from the effective Lagrangian of Euler *et al.* [16]. Straight lines are electrons; wavy lines are photons.

Euler *et al.* used this effective Lagrangian only in the tree approximation, to calculate the leading terms in photon interaction matrix elements. It was not until much later that such Lagrangians, though non-renormalizable, were used beyond the tree approximation [12,17].

In modern jargon, we say that in deriving this Lagrangian the electron is 'integrated out', because in the one-loop approximation we have

$$\exp\left(i\int \mathcal{L}_{\rm eff}(\mathbf{E},\mathbf{B})\,d^4x\right) = \int\left[\prod_x d\psi_e(x)\right]\exp\left(i\int \mathcal{L}_{\rm QED}(\psi_e, A)\,d^4x\right).$$

A more general procedure is simply to write down the most general non-renormalizable effective Lagrangian, use it to calculate various amplitudes as an expansion in energies and momenta, and then choose the constants in the effective Lagrangian by matching the results it gives for these amplitudes to those derived from the underlying theory.

We will encounter effective field theories again, especially in considering broken symmetries in Volume II. As we shall see, effective field theories are useful even where they cannot be derived from an underlying theory, either because the theory is unknown, or because its interactions are too strong to allow the use of perturbation theory. Indeed, even if we knew nothing about the properties of charged particles, the scattering of photons at sufficiently low energy would have to be described by an effective Lagrangian consisting of the terms $(\mathbf{E}^2 - \mathbf{B}^2)^2$ and $(\mathbf{E}\cdot\mathbf{B})^2$, because these are the unique quartic Lorentz- and gauge-invariant terms with no derivatives acting on $\mathbf{E}$ and $\mathbf{B}$. Terms with such derivatives would be suppressed at low photon energies E by additional factors of E/M, where M is some typical mass of the charged particles that are being integrated out. We can go further: we shall see that effective field theories are useful even where the light particles they describe are not present in the underlying theory at all, but composites of the heavy particles that are integrated out. The underlying theory might not even be a field theory at all—the problem of incorporating gravitation has led many theorists to believe that, in fact, it is a string theory. But wherever an effective field theory comes from, it is inevitably a non-renormalizable theory.